

X'Presso - Complete Feature List (Consolidated)

This document lists all product modules and features for the X'Presso platform. Nothing omitted from the agreed consolidated scope.

1) Identity & Access (Core Platform)

- Identity management
- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-based access control (student, faculty, admin, placement officer, etc.)
- Directory services
- Identity governance (joiner-mover-leaver, access reviews, audit)

2) Admissions & Institution Onboarding

- Admissions and application management
- Applicant status workflow: applied, shortlisted, offered, accepted, enrolled, active
- Admission-to-institution onboarding at enrollment
- Auto-create and link student to institution during enrollment
- Assign program, batch, semester, and enrollment status
- Generate institutional identity (student ID, username or email per policy)
- Trigger onboarding actions (portal access, welcome notifications, defaults)

3) Student & Academic Core (IMS)

- Student Information System: profiles, records, transcripts
- Enrollment management
- Course registration
- Degree planning and academic advising
- Timetable and class scheduling
- Curriculum management
- Assessment and accreditation support

4) Attendance Management (V1)

- Faculty attendance marking (manual, bulk, timetable-based)
- Student attendance visibility (course-wise, period-wise)
- Attendance rules (thresholds, condonation, leave and late handling)
- Low-attendance alerts (student, advisor, admin)
- Attendance-to-eligibility linkage (exam, internal, placement)
- Correction workflow with full audit trail

5) Teaching & Learning (ALS)

- LMS and course runtime (courses, modules, lessons)
- Assignment management
- Grading and gradebook
- Online assessments and exams
- Live virtual classes
- Lecture capture
- Digital content and material management
- Adaptive learning pathways

- Learning analytics (engagement, mastery, outcomes)
- Student success alerts and interventions
- AI tutor and coach (course-aware)

6) Placement & Career Services (V1)

- Student-facing placement portal (always accessible from student portal)
- Admin posting and management of job openings
- Targeted visibility (program, batch, eligibility)
- Student applications and tracking
- Status flow: applied, shortlisted, interview, offer, rejected
- Resume and profile management
- Placement notifications (in-app, email, SMS)
- Placement analytics and export reports

7) Education and Knowledge Management (EMS)

- Research information management
- Grant proposal and funding management
- Research finance tracking
- Lab and experiment management
- Research ethics and compliance workflows
- Institutional repository (papers, datasets, code)
- Research data management (metadata, versioning, lineage)
- Knowledge graph (people, papers, datasets, courses)
- Publishing workflows (preprint, journal, IP)
- Discovery and search (semantic, cross-discipline)
- Collaboration (intra- and inter-university)

8) Enterprise Operations (ERP and Admin)

- Finance and accounting
- Procurement and vendor management
- Budget planning
- Expense management
- Financial reporting and compliance
- HR management
- Recruitment and hiring
- Performance management
- Faculty information system (teaching load, research output, tenure)
- Workflow and BPM automation
- Enterprise content and records management

9) Campus and Institutional Operations

- Facilities management
- Room scheduling
- IT service desk and ticketing
- Library operations (access, licensing)
- Campus card and access control integration
- Physical security integration

10) Data, Analytics and Intelligence (Core Platform)

- Institutional data warehouse or lakehouse
- Business intelligence dashboards
- Institutional reporting and accreditation reports
- Master Data Management (MDM)
- Outcome-first dashboards (skills, competency, outcomes)
- Cross-module analytics (academics, operations, placements, research)

11) Integration and API Platform (Core Platform)

- API gateway and management
- Integration middleware (ESB or iPaaS style)
- Event streaming for real-time data flow
- Connector framework (SIS, ERP, LMS, video, payment)
- Developer and integration portal (later phase)

12) Infrastructure, Security and Reliability (Core Platform)

- Cloud and infrastructure management
- Network and system operations visibility
- Security operations and SIEM integration
- Observability (logs, metrics, traces)
- Backup and disaster recovery
- Endpoint and device management hooks

13) Communication and Engagement (Cross-cutting)

- Announcements
- In-app notifications
- Email and SMS notifications
- Audience targeting (role, program, batch)
- Communication audit and logging

14) Compliance, Trust and Quality (Cross-cutting)

- Full audit logs for critical actions
- Data retention and records policy controls
- Consent and privacy controls
- Accessibility baseline (WCAG-oriented UX)
- Export and report packs for institutional audits